Abstract

Source: `paper/abstract.tex` *at commit* `e6e12cd`*. The Japanese manuscript is authoritative. The abstract is short, so this part renders it sentence by sentence rather than summarizing it.*

This paper formulates the autonomous cognitive system (ACS) as an object for describing, in a common form, systems that distinguish themselves from their environment, select actions in response to inputs, undergo internal state transitions, and update the state of their internal model while seeking to persist.

As a foundation for the Ethical Coordinative Intelligence Theory (ECIT), which the author is developing, the theory aims to provide a preliminary framework for the formal study of ethical coordination from the standpoint of feasibility preservation.

The formal content of this paper is restricted to Phase 1, a deterministic, discrete-time framework that assumes compactness of the state space, the action space and the internal-model state space. It centres on the internal dynamics of a single ACS. Organizational ACSs are treated as a limited description that applies the single-ACS framework structurally to social organizations.

An ACS is defined as a 7-tuple $\Sigma = (X, U, E, f, \Pi, M, \mathrm{Update})$. The feasible set $K := \{x \in X \mid \Pi(x) \geq 0\}$, derived from the persistence evaluation function Π, is the central object of the description of feasibility.

Three principles are introduced as constraint axes bearing on the maintenance of long-term feasibility: self-maintenance (SM), environmental utilization (EU) and feasible response (FR). For each principle, its time-indexed satisfaction predicate and its logical status are set out and kept distinct. FR is the constraint axis concerned with a feasible response being obtainable when one is needed, and with the retention, acquisition, recovery and reconstruction of the potential for such needed responses.

The three principles are treated as constraint axes at the principle level. Separately, as a description at the mechanism level, the paper gives an internal mechanism structure. It consists of the Triadic Information-Processing Architecture (TIPA), which carries out internal evaluation bearing on feasibility and the formation of responses; the Physical Substrate Layer (PSL), which supports the operation of TIPA; and the Implementation Substrate Layer (ISL).

The paper also formulates a standard operation form, in which the record of one step of operation closes into the next time step, and the connection between time steps. It further formulates the minimal structure of relations with external objects, and relation quality.

The paper distinguishes *blockage*, in which no feasible action exists, from *collapse*, in which the organizational relations that had constituted the system's identity as a structural individual are lost in such a way that they are not recovered as the same individual.

It also distinguishes the groups and organizations in which ACSs participate from the organization's own status as an ACS. Where a well-formed ACS structure is realized for the organization itself, the organization is described as an organizational ACS, whether or not individual ACSs participate in it, and the definitions of the preceding sections are applied at the organizational level.

ACS theory aims at a common form for describing the actions of different autonomous agents, prior to any particular cultural judgment of good and bad, as structural relations to feasibility requirements, internal states, the external environment, relations, and the operation of internal mechanisms. This paper gives its deterministic, structural foundation.

The paper takes the standpoint of a descriptive and structural theory. It does not claim, directly from its formal results, empirical validity for humans, other living organisms, AI systems, robots, organizations or the like.

Auxiliary refinement within the deterministic framework is deferred to Phase 1.α. Stochastic extension, continuous-time formulation, causal relational dynamics, temporal self-identity that permits branching, and the quantification of social dynamics are deferred to Phase 2. The formal connection with ECIT is deferred to Phase 3.

Keywords: autonomous cognitive system; feasibility; viability theory; persistence evaluation function; three principles (self-maintenance, environmental utilization, feasible response); feasibility-preservation constraint; TIPA; organizational ACS; blockage; collapse.